

NILAY SINGH

+91 7982631261 ✦ Gurugram,INDIA
nilay31singh@gmail.com ✦ [LinkedIn](#) ✦ [Github](#)

EDUCATION

B.Tech in Computer Engineering, Thapar University
CGPA: 8.85

Aug 2023 - May 2027

SKILLS

Languages	Python, JavaScript, HTML/CSS, SQL
Frameworks	React, FastAPI, Streamlit, LangChain, Pandas, Numpy ,Matplotlib
AI / ML & Tools	Machine Learning (CatBoost), RAG, LLMs (Gemini, Ollama), FAISS, Git

PROJECTS

RAG-CHATBOT: Local AI Document Chat Application

[\[GitHub Link\]](#)

An enterprise-ready solution ensuring **100% data privacy** for AI-powered document analysis.

Tech Stack: Python, FastAPI, React, Ollama (Phi-3), FAISS, Tailwind CSS, JavaScript

- Powered a real-time, conversational Q&A experience by architecting a **Retrieval-Augmented Generation (RAG) pipeline** in **FastAPI**, leveraging a **FAISS vector database** for near-instantaneous semantic search.
- Delivered an intuitive and seamless user experience by developing a **responsive chat interface** with **React** and **Tailwind CSS**, connected via **RESTful APIs**.
- Guaranteed **100% data privacy** and offline functionality by integrating a **locally-run LLM** (Phi-3 via Ollama), eliminating the need for third-party API calls.

Mental Health Predictor & EDA Dashboard

[\[GitHub\]](#) [\[Live Demo\]](#)

A data-driven tool to uncover mental health trends and identify individuals who may **benefit from support**.

Tech Stack: Python, Streamlit, CatBoost, Pandas, Plotly, Seaborn, Matplotlib

- Demystified complex mental health data by creating an **interactive dashboard** with **Streamlit** that integrates an **EDA module** using **Plotly** and a quiz-based prediction interface.
- Constructed a highly accurate predictive engine by implementing and training a **CatBoost classifier** on tech industry survey data to identify individuals needing support.
- Maximized the model's real-world effectiveness by tuning parameters to achieve a **Recall of 0.83** and an **F1-Score of 0.74**, ensuring reliable identification of at-risk individuals.

SkillEvalAI: AI-Powered Career Readiness Analyzer (In Progress)

[\[GitHub Link\]](#) [\[Live Demo\]](#)

An AI-powered assistant that evaluates tech career readiness and builds a 30-day personalized learning roadmap.

Tech Stack: React, Tailwind CSS, Vite, FastAPI, LangChain, Gemini LLMs, GitHub API

- Empowered** aspiring tech professionals to assess job readiness and bridge critical skill gaps **by developing** a comprehensive AI assistant that generates detailed readiness scores and identifies specific skill deficiencies for various tech roles.
- Enabled** core intelligent functionalities and personalized guidance **by engineering** a robust backend with **FastAPI** and **LangChain**, integrating **Gemini LLMs** to perform advanced profile summarization, in-depth skill gap analysis, and dynamic 30-day learning roadmap generation.
- Ensured** holistic, data-driven profile evaluations **by seamlessly integrating** the **GitHub API** to process extensive user coding footprints, project details, and skills for highly accurate and actionable assessments.
- Enhanced** user engagement and accessibility **by designing and implementing** a modern, intuitive, and responsive frontend using **React** and **Tailwind CSS**, providing instant feedback and a streamlined user experience.

CERTIFICATIONS

Machine Learning Specialization

[\[View Credential\]](#)

July 2025

Issued by **DeepLearning.AI & Stanford University**

- Pioneered** advanced machine learning applications, encompassing **supervised learning** for predictive modeling, **deep neural networks** with **TensorFlow**, and the development of **reinforcement learning** models for complex AI systems.